

A roton in the emergent photon spectrum of quantum spin ice

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Quantum spin ice is expected to host an emergent photon, and every analytic prediction of what neutrons should see treats that photon as free: one sharp line carrying all of the spectral weight. We compute the spectrum exactly instead, diagonalizing the ring-exchange model in the full spin-ice configuration space of 48- and 64-site pyrochlore clusters. The free-photon picture holds at long wavelength and fails at the zone boundary, where a *roton* takes over: at the L point a mode falls to half the free-photon energy, carries half of the neutron intensity, and is the lowest excitation of the model. Its origin is the one Feynman identified in liquid helium, with a single ingredient replaced. In helium a sound wave becomes a roton because the liquid can no longer be compressed and must rearrange instead. Here the emergent electric field is the spin itself, $E = S^z = \pm \frac{1}{2}$ on every bond, so the field is saturated: a wave cannot be made by increasing E , only by rearranging which bonds point which way. That rearrangement twists the magnetic flux and destroys ring-exchange coherence, and the excitation energy is exactly the coherence destroyed. Plaquettes lying in the wavefronts enclose no flux from the wave and cost nothing, and at L alone — the one zone-boundary momentum along a hexagon normal — a quarter of all plaquettes are of this type. Two further consequences of compactness accompany the roton: the two photon polarizations, exactly degenerate in the free theory, are split by their mutual interaction, and 13–38% of the neutron weight moves into multiparticle states. A selection rule — reversing all spins leaves the Hamiltonian invariant, and neutrons see only the odd states — sends the complementary even spectrum to Raman scattering and ultrasound. The zone-boundary feature experiments should expect is therefore not the photon but the roton.

Introduction.—In quantum spin ice the low-energy physics is not made of spins but of a field. Virtual exchange on the two-in–two-out ice manifold of the pyrochlore lattice [1] flips the six spins around a hexagon, and the resulting theory is a compact $U(1)$ lattice gauge theory with an emergent photon [2–6], confirmed by quantum Monte Carlo [7, 8] and realized equivalently by three-dimensional quantum dimer models [9]. The dictionary is simple and exact: the local moment is the emergent electric field, $S^z = E$, so the neutron structure factor $S^{zz}(\mathbf{q}, \omega)$ measures electric-field fluctuations. A gapless linearly dispersing mode carrying little low-energy weight is the signature now pursued in a growing family of materials [10–19].

Every analytic prediction of that signature describes a *free* photon, and the approximation behind it is stated openly in the literature. The emergent magnetic field is an angle, so its energy is a cosine; one assumes that averaging over fast fluctuations both softens the discreteness of the spin variable and keeps the flux small, expands the cosine to quadratic order, and arrives, in the words of Benton *et al.*, at “the Hamiltonian for non-compact quantum electromagnetism” [2, 10]. Later treatments of the neutron response inherit it [20, 21]. The approximation is a statement about long wavelengths, and there it is justified: in three spatial dimensions the deconfined phase does flow to free electrodynamics, monopoles are gapped, and the corrections are irrelevant.

At the zone boundary none of that protection applies, and the reasons to worry are concrete: the emergent fine-structure constant is large [22], semiclassical treatments

find order-one anharmonic renormalizations [23–25], and quantum Monte Carlo finds a zone-boundary linewidth comparable to the whole photon bandwidth, explicitly “not captured” by the Gaussian theory [26]. Monte Carlo cannot say what replaces the photon, because recovering a spectrum from imaginary time smears exactly the structure in question. That is the gap we close. The zone boundary is not a corner case: it is where the neutron-accessible momenta lie.

The model, and why nothing is missing from it.—We study the ring-exchange Hamiltonian on the ice manifold,

$$H = -K \sum_p (W_p + W_p^\dagger), \quad W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-, \quad (1)$$

summed over contractible hexagons. W_p reverses the six spins around a hexagon when they alternate and annihilates the state otherwise, so H rewards *resonance*: a configuration gains energy by being connected, through flippable hexagons, to its neighbors. In gauge language $W_p = e^{iB_p}$ and Eq. (1) is $-2K \sum_p \cos B_p$, the full cosine, never expanded. Every difference from the free-photon theory below comes from that one refusal.

Compact electrodynamics is normally written $\frac{U}{2} \sum_\ell E_\ell^2 - K \sum_p \cos B_p$, and Eq. (1) appears to be missing the electric term. It is not. For spin one-half the electric field is the spin, $E_\ell = S_\ell^z$, so $E_\ell^2 = \frac{1}{4}$ on *every* bond of the ice manifold: there the electric term is a constant and generates no dynamics. Equation (1) is the whole gauge Hamiltonian at this order. This is worth saying plainly, because it is the physical reason the problem is hard: spin-one-half quantum spin ice

is compact $U(1)$ gauge theory with *zero* bare electric stiffness, the extreme of strong coupling, and whatever stiffness the photon has is generated by the ring dynamics itself [2, 10, 23]. There is no small parameter to tune. (For spin $S > \frac{1}{2}$, or in rotor and quantum-link regularizations, E^2 is dynamical and the coupling can be made weak; the spin-one-half magnet cannot be.)

Equation (1) conserves the winding of the spin configuration around the periodic box, contains no spinons — the Hilbert space is the ice manifold — and excludes the noncontractible six-cycles responsible for torus artifacts in finite-cluster calculations [27]. It does retain the full magnetic sector, so weight beyond the photon could in principle be magnetic rather than multi-photon; we test that below.

One operator does everything.—Because $S^z = E$, the neutron operator is also the photon creation operator, and this single fact organizes what follows. In the free theory $E(\mathbf{q})|0\rangle$ is one photon, exactly, which is why that theory places all of the weight on one line. In the compact theory the same state is a superposition of everything the probe can reach, and our question is what that superposition contains.

The two fields have a plain meaning in spins, and we use both languages throughout. The electric field is the moment itself, $E_\ell = S_\ell^z$, so an electric-field wave is a modulation of the up–down pattern. The magnetic flux B_p threads a hexagon, and $\cos B_p$ is the ring-exchange resonance: a hexagon with a large $\langle \cos B_p \rangle$ is one whose six spins are flipping back and forth coherently. The ground-state energy is the sum of those resonances.

What replaces the photon.—Figure 1 shows the exact S^{zz} beside the free-photon prediction evaluated on the same lattice, so that every difference is an effect of compactness alone. Two diagnostics classify the levels. Projecting the probe onto the two transverse polarization directions of the free theory says which photon branch a level belongs to. The second uses the one other coupling this model is conventionally given. Write

$$H(V) = -K \sum_p (W_p + W_p^\dagger) + V \sum_p n_p^{\text{flip}}, \quad (2)$$

where K is the ring-exchange amplitude, which sets every energy quoted here, and n_p^{flip} is the projector onto hexagons whose six spins alternate — the flippable ones — so that $\sum_p n_p^{\text{flip}}$ simply counts them. The V term is the Rokhsar–Kivelson potential [28], the standard second axis of the quantum-ice phase diagram [3, 7], and at $V = K$ it reaches the exactly solvable point where the ground state is the equal-amplitude superposition of ice configurations. We use it only as an infinitesimal probe at $V = 0$, because the Hellmann–Feynman theorem makes its slope an exact census:

$$\frac{d\omega_n}{dV} = \langle n | \sum_p n_p^{\text{flip}} | n \rangle - \langle 0 | \sum_p n_p^{\text{flip}} | 0 \rangle, \quad (3)$$

the number of flippable hexagons the excitation adds or removes relative to the ground state. It is a measurement of what the state is made of, not a fitted parameter.

At L the two dominant levels behave completely differently. One sits at $1.73K$, near the free-photon line at $2.03K$, and loses flippable hexagons under the deformation: an ordinary dressed photon. The other falls to $1.02K$, half the free-photon energy, takes 51% of the neutron intensity, *gains* flippable hexagons, and is the lowest excitation of the entire model — we scanned all 9104 levels of the ground sector, labeled by spin-reversal parity, and nothing of either parity lies beneath it. It is not two photons or three; those are excluded by parity and by kinematics, and the flux-correlation test below excludes a magnetic pair. It is a roton, and the rest of this paper is about what that means microscopically.

What an electric-field wave costs.—Electric and magnetic variables are conjugate, so an electric-field wave necessarily disturbs the flux, and the ring exchange is the entire energy of the problem. The cost is therefore magnetic, and for Eq. (1) it can be written down in closed form. The first moment of the structure factor is

$$f_1(\mathbf{q}) = \frac{1}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu |g_p^\mu(\mathbf{q})|^2, \quad (4)$$

where $g_p^\mu(\mathbf{q})$ is the lattice curl — the flux the wave drives through hexagon p . Equation (4) is exactly the Maxwell magnetic energy $\sum_p (\text{stiffness}) \times |\nabla \times A|^2$ of the configuration the neutron creates, with one difference from the usual lattice gauge theory: the stiffness is not a parameter but a measured property of the ground state, $\langle W_p + W_p^\dagger \rangle = 2\langle \cos B_p \rangle$, the strength with which that hexagon is already resonating. This is the stiffness that free-photon treatments must introduce by hand [2, 10]; here it is computed. Equation (4) contains no free parameter, reproduces the exact spectral first moment at every momentum, and obeys $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$ identically. It plays the role that the free-particle recoil $\hbar^2 q^2/2m$ plays in helium.

In spin language the same statement reads: twisting the flux through a hexagon spoils the coherence of the six-spin resonance there, and the energy lost is the resonance strength times the square of the twist.

The energy of the state the neutron creates is then $f_1(\mathbf{q})/S(\mathbf{q})$ [29], where the equal-time structure factor $S(\mathbf{q})$ measures how much of that electric-field pattern the ground state already contains: where the field is already fluctuating that way, exciting it costs less. *The magnetic cost, divided by what was already there* — that is the whole argument, and it is the argument Feynman used for the helium roton.

Why at L .—Two things happen there at once, one geometric and one dynamical [Figs. 1 and 2].

The geometric one is exact, and it is ordinary electromagnetism on a lattice. A hexagon lying *in* a wavefront

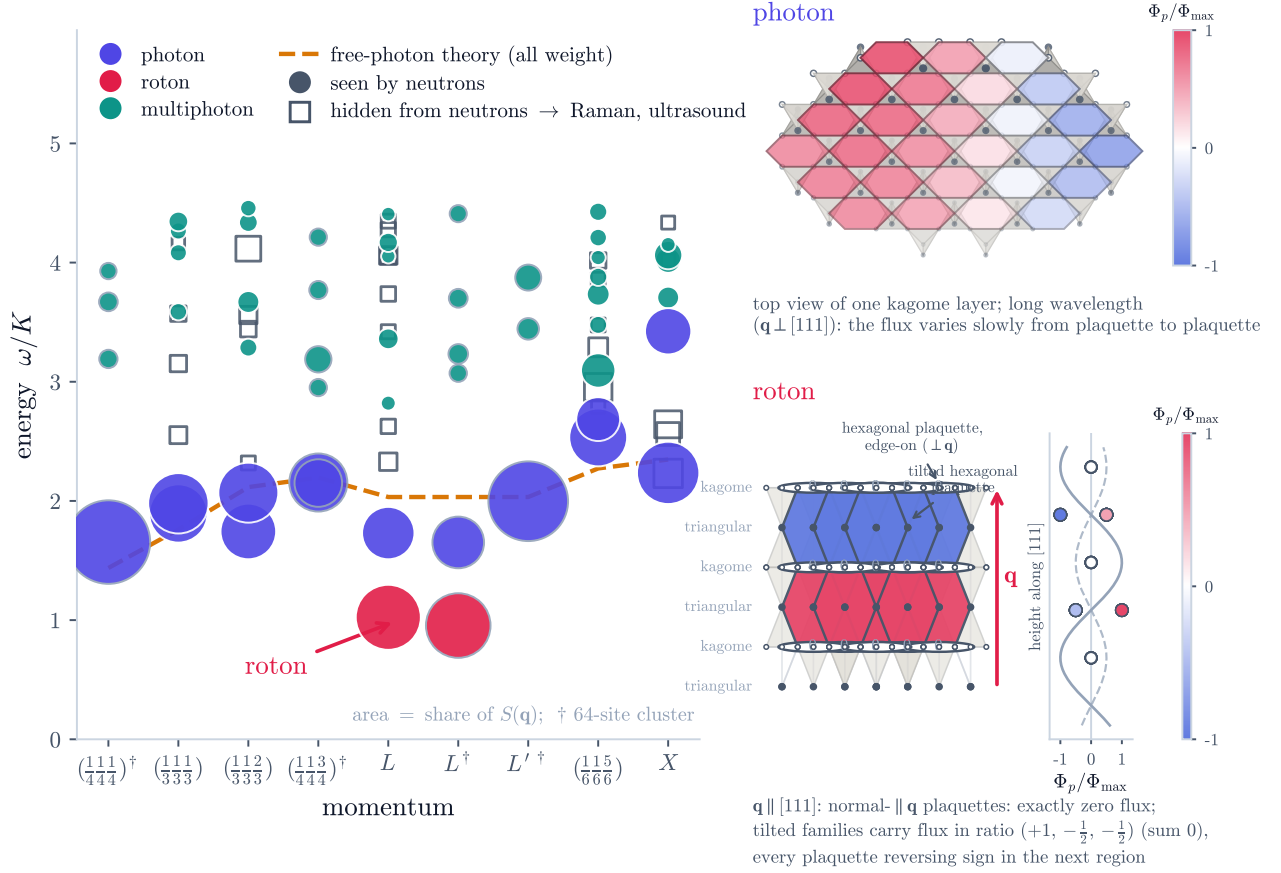


FIG. 1. The exact excitation spectrum of the compact gauge theory, and what each excitation is. Left: every level carrying spectral weight at nine inequivalent momenta, ordered by $|\mathbf{q}|$ (48-site cluster; $\dagger$ marks momenta of the 64-site cluster). Marker area, and every percentage quoted in this paper, is the level's *share* of the total neutron intensity at that momentum, $|\langle n|S^z(\mathbf{q})|0\rangle|^2/\sum_m |\langle m|S^z(\mathbf{q})|0\rangle|^2$; open squares mark levels invisible to neutrons that appear instead in Raman scattering and ultrasound — by the parity selection rule no level is seen by both. Colors state the character established by the diagnostics of the text, not by position: *photons* are the two levels that project onto the transverse polarizations of the free theory, sit within 0.82–1.46 of the free-photon energy, and do not gain flippable hexagons; the *roton* is the level that is anomalous on both counts at once — half the free-photon energy, and the only level in the spectrum that appreciably *gains* flippable hexagons (+0.34, against at most +0.01 for every other bright level) — and which the flux diagnostic distinguishes from a monopole pair; *multiphoton* is the remaining neutron-visible weight, absent from the free theory at any parameters. At daggered momenta the assignment follows the dominant line and the class analysis of the text. The dashed curve is the free-photon theory on the same lattice, which puts *all* of the weight on that single line. Three L -type columns test the roton directly. It appears on both clusters — at L (48 sites: 1.02K, 51% of the intensity) and at $L^\dagger$ (64 sites: 0.95K, 50%) — while L' , the second L -type class of the anisotropic 64-site box, whose ground-state structure factor is suppressed, does not rotonize: rotonization follows $S(\mathbf{q})$, as the mechanism requires. Right: what each excitation is, drawn on the pyrochlore lattice with every plaquette colored by the flux Φ_p the wave drives through it (colorbars). A *photon* is a long-wavelength wave, here $\mathbf{q} \perp [111]$: the flux varies slowly from plaquette to plaquette. The *roton* is the shortest wave along $[111]$, shown in a side view with $[111]$ vertical, the alternating kagome and triangular planes labeled, and the layer spacing exaggerated: the kagome plaquettes appear edge-on as white slats and enclose *exactly* zero flux, because their normal is parallel to $\mathbf{q}$; within one interlayer region the three tilted families carry flux in the exact ratio $(+1, -\frac{1}{2}, -\frac{1}{2})$ — summing to zero, as a divergence-free flux pattern must — and every tilted plaquette's flux reverses sign in the next region. The lattice drawing shows the strongest family; the profile plots Φ_p against height for all three (solid and dashed guides). Flux is quoted in units of its maximum; the overall amplitude is the mode's and is arbitrary here. The multiphoton levels (teal) are weight with no free-theory counterpart: they exist because the cosine carries the quartic vertex $-\frac{K}{12}B_p^4$ that the free theory discards, which lets one photon become three.

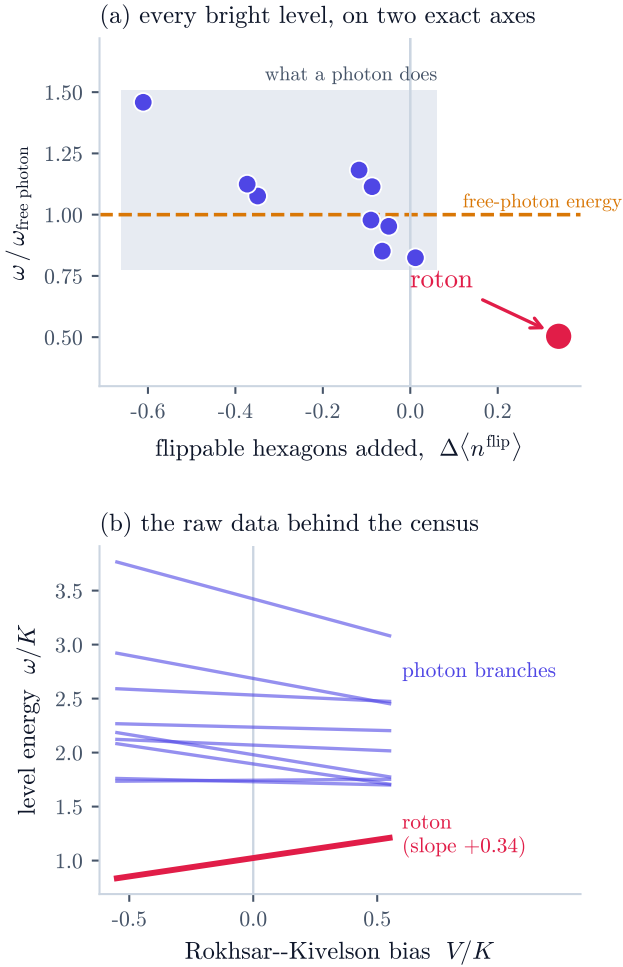


FIG. 2. The sharp diagnostic is two-dimensional. (a) Every bright level of the 48-site spectrum placed on two exactly computed axes. Vertical: the energy in units of what the *free-photon* theory assigns to that same momentum, so a photon must sit near 1. Horizontal: $\Delta \langle n^{\text{flip}} \rangle$, the Hellmann–Feynman census of Eq. (3) — how many flippable hexagons the excitation adds or removes. The photon branches fill the shaded box: 0.82–1.46 of the free-photon energy, and never appreciably gaining hexagons (at most +0.01). The roton lies outside on *both* axes: half the free-photon energy, and +0.34. Neither axis alone separates it — one photon-branch level also has a marginally positive census, another also lies below the free-photon line — but no other level is anomalous in both. (b) The data behind the horizontal axis: each level tracked through the bias V . Census values are evaluated exactly at $V = 0$; the finite-difference slopes agree with them to ~ 0.01 except where levels cross, which is why the census, not the fit, defines the axis.

— in a plane of constant phase, perpendicular to $\mathbf{q}$ — encloses no net flux from the wave: the curl g_p vanishes identically, because the six sites sit at one phase and the alternating signs of the loop cancel. Such plaquettes are threaded by nothing, so by Eq. (4) they contribute no magnetic energy, and their resonance is left intact. Each

pyrochlore hexagon is perpendicular to one of the four $\langle 111 \rangle$ axes, and L is the only zone-boundary momentum lying along such an axis. There, one family of hexagons in four lies in the wavefronts and drops out of the cost exactly, reducing f_1 to three quarters of its value at neighboring zone-boundary momenta. A wave can be built at L while a quarter of the plaquettes never notice it.

The dynamical one is that $S(\mathbf{q})$ is largest at L (0.370 at L against 0.19–0.29 elsewhere). A smaller cost divided by a larger head start makes f_1/S minimal at L among all twelve cluster momenta, and L is the only momentum where it falls below the free-photon line.

Neither effect is inherited from the ice rule. Sampling the classical ice manifold directly, on a 2048-site pyrochlore, gives a structure factor flat across the zone to within a few percent, and the equal-amplitude ice state on our own cluster favors L by only about ten percent. The interacting ground state concentrates far more weight there while *suppressing* other momenta. Unlike the helium roton, this one is not a property of the medium it lives in; the ring dynamics manufactures it.

What the roton is made of.—The exact state answers this directly, and the answer is the same in either language. Relative to the ground state it carries *less* magnetic coherence: $\sum_p \langle \cos B_p \rangle$ falls, and the amount it falls by, $1.02K$, is exactly the excitation energy. The flux has spread away from $B_p \simeq 0$. In spins, the six-spin resonances have gone out of phase with one another. At the same time the state contains *more* hexagons in a flippable arrangement (+0.34). The two facts are not in conflict — how many plaquettes *can* resonate and how coherently they *do* are different questions — and together they say what the roton is: a rearrangement of the electric-field pattern that leaves more plaquettes able to fluctuate while destroying the phase agreement among them. It pays in coherence, not in ice-rule strain, which is why it costs half of what stiffening the flux into a photon costs.

A second stage happens where the neutron cannot look. The bare electric-field wave costs $1.45K$; the true roton costs $1.02K$, so the system recovers 29% of the price. Resolved plaquette by plaquette, 71% of that recovery comes from the one family lying in the wavefronts. The reason is that those plaquettes carry none of the wave: their flux can be rearranged *without* reducing the state’s overlap with what the neutron created, so they are free to be spent. The wave is built on three families of plaquettes and paid for largely by reorganizing the fourth. This is the lattice counterpart of backflow in helium [30], where the liquid recovers energy through motion the density wave does not specify, and it is why the exact energy lies well below f_1/S [Fig. 2(a)].

Is this not simply another photon?—Genealogically, it is: the polarization decomposition connects the level continuously to the second transverse branch of the free theory, and no new particle appears from nowhere. The situation is exactly that of helium, where the roton is not a

separate species but the high-momentum portion of the very branch that starts as sound — one dispersion curve whose *character* changes along the way. “Renormalized photon,” however, would mean the same character with shifted numbers, and that reading fails every test that probes character rather than ancestry. A photon’s energy *falls* under the flippability bias; the roton’s *rises* — a sign flip, not a shifted magnitude [Fig. 2(b)]. A photon pays its energy in flux stiffness while preserving ring-exchange coherence; the roton carries *more* flippable plaquettes with *less* coherence. A photon’s energy is set by the generated stiffness; the roton’s is set by the equal-time structure factor through f_1/S , tracking $S(\mathbf{q})$ class by class on the 64-site cluster and point by point along the deformation. And no photon detaches from its band: the roton sits below the entire photon spectrum and is the lowest excitation of the model. The name marks a change of character, not of ancestry, and the change is measured in sign flips, not percentages.

Relation to known soft modes.—The closest antecedents are the “resonons” of quantum dimer models. At the fine-tuned Rokhsar–Kivelson point the ground state is an equal-amplitude sum, its correlations are those of the *classical* dimer ensemble, and the same single-mode logic then predicts soft modes wherever those classical correlations peak [28]: on the square lattice, soft resonons at (π, π) and $(\pi, 0)$, confirmed dynamically [31]. In three dimensions the same analysis finds nothing of the kind: at the cubic-lattice RK point the only soft mode is the remnant of the photon itself [31], because the classical Coulomb-phase correlations are featureless — the same flatness we measure for the classical ice manifold. The roton reported here therefore has no RK-point ancestor. It lives deep in the deconfined phase with no fine-tuning, at finite energy rather than at a critical zero, and it exists only because the quantum ring dynamics manufactures the correlation peak that the classical ensemble lacks. Where the resonon is a critical precursor read off from classical correlations, this roton is an interaction effect of the deconfined gauge field itself.

Why compactness is what makes this happen.—In the free theory E is a continuous field and a photon is a small-amplitude oscillation of it: the electric field grows a little, the flux stiffens a little, and the two exchange energy indefinitely. In the magnet $E = S^z = \pm \frac{1}{2}$ on every bond, so the field is *saturated*. A wave cannot be made by increasing E anywhere, because there is nowhere for it to go; it can be made only by rearranging which bonds carry which sign, and that rearrangement is exactly what changes the flux and costs the resonance. At long wavelength the rearrangement can be spread over many bonds and is indistinguishable from a small oscillation — which is why the free photon survives in the infrared and why the multiparticle weight falls toward small $\mathbf{q}$. At the zone boundary the two part company, and the rearrangement is the cheaper option. *The roton is what an electric-field*

wave becomes when the field can no longer change its magnitude, only its arrangement. In helium the binding constraint is that the liquid cannot be compressed below the interparticle spacing; here it is that a spin one-half cannot exceed one-half. The constraint differs; the consequence is the same.

Evidence that this is one mode and not an accident.—Three checks. First, the mode is collective: improving the electric-field wave variationally — the systematic version of adding backflow — converges onto the exact energy within a few steps at L , faster than at any other momentum ($1.450 \rightarrow 1.074 \rightarrow 1.034 \rightarrow 1.026K$, against $1.024K$ exact). Second, the mechanism predicts *which* momenta rotonize. On the 64-site cluster the L -type momenta split into two inequivalent classes, and the class with the larger $S(\mathbf{q})$ develops the low mode carrying half the intensity while the other stays photon-dominated: rotonization follows the structure factor, not the label L . Third, it predicts how the mode moves. Deforming toward more flippable states leaves f_1 constant to under a percent while $S(L)$ grows, and the roton softens monotonically as f_1/S requires [Fig. 2(b)], from $1.02K$ to $0.73K$ without condensing in the range scanned. With f_1 fixed, the softening is not merely correlated with the growth of S ; it is caused by it.

The other two consequences of compactness.—Compactness does two more things visible in Fig. 1. The free theory makes the two transverse polarizations exactly degenerate at every momentum of this cluster; the exact spectrum splits them, by under a percent at the smallest momentum and by order one at the zone boundary. Because the degeneracy is exact in the free theory, the splitting measures the photon–photon interaction directly. And beyond both branches, 13–38% of the neutron weight sits in higher states that the free theory does not contain at all. Their character is multiparticle rather than magnetic: the plaquette flux correlations of every bright state are weak and spatially spread out, without the localized disturbance a bound monopole pair would produce, and their response to the deformation matches sums of single-mode responses. This weight falls to 13% at the smallest momentum reachable, as the recovery of the free-photon description at long wavelength requires.

Leading-order perturbation theory in the quartic term of the cosine accounts for none of this. Evaluated on the same modes, it predicts polarization splittings and transferred weight below 0.1% at the coupling this model generates, and a few percent even with the coupling tripled — one to two orders of magnitude short. The photon–photon problem here is not a perturbation of the free theory.

A rule that divides the labor between experiments.—Reversing every spin leaves Eq. (1) unchanged, so every eigenstate is even or odd under it, and S^z , being odd, connects the ground state only to odd states. (In gauge language this is charge conjugation, and an n -photon state

carries parity $(-1)^n$.) The rule is exact, and we confirm it at machine precision: every even level carries neutron weight below 10^{-28} . Its consequence is practical. The two-photon continuum, and the even half of the spectrum generally, is *invisible* to neutrons and appears instead in probes coupling to pairs of spins — Raman scattering [32] and ultrasound [33]. Figure 3 shows both channels; no level appears in both. A bilinear-channel onset at twice the neutron line would identify the emergent photon pair with no adjustable parameter. Stray-field noise magnetometry [34] couples to the magnetization and so sits on the neutron side of the division.

Consequences for experiment.—Free-photon fits to neutron data are least reliable exactly where the accessible momenta lie. At the zone boundary between a sixth and two thirds of the weight is not in the one-photon line; at L the photon retains only a third of it; and the line itself is split. The dominant zone-boundary feature is the roton, at half the energy the free theory assigns to the photon there, so a measured low-energy peak should not be read as a photon whose speed has been renormalized. The zone-boundary damping reported by quantum Monte Carlo [26] acquires a concrete assignment: weight moved into the split doublet and the multiparticle states. Because the theory is gapless, multi-quantum continua reach down to the one-photon line in the thermodynamic limit, so lineshape rather than peak counting is the appropriate analysis.

Limitations.—Two cluster sizes. The roton’s energy, its share of the weight, and the mechanism are stable between them; the splitting at X is not, and we do not build on it. We locate the roton at the momenta the clusters provide but do not predict its position throughout the zone: that requires $S(\mathbf{q})$ of the interacting ground state at momenta these sizes cannot reach, and the classical ice manifold cannot be substituted for it. Since Eq. (4) holds at any $\mathbf{q}$, that one ingredient would turn the mechanism into a zone-wide prediction. Longer ring exchanges present in microscopic models lie outside Eq. (1). The connection to those models is not made here: a companion analysis of the XXZ model with torus artifacts removed [27] finds comparable off-pole weight at L , but there the photon-band states carry virtual-spinon admixture and the comparison is deferred; the overall scale K can be targeted thermodynamically [35].

Choice of cluster.—One technical point deserves recording, because it affects any small-cluster study of this problem. The standard 16- and 32-site pyrochlore cells cannot be used. On the 32-site cell every configuration of the connected zero-flux sector has exactly eight flippable hexagons, which makes the equal-amplitude state an exact eigenstate and the spectrum integer-valued; on the 16-site cell only six transport sectors have any dynamics at all. Neither supports the interacting photon problem. We therefore use the $2 \times 2 \times 3$ cluster (48 sites, 87 546 ice states, ground sector of dimension

9104, diagonalized completely, so the weights saturate the equal-time correlator identically) and, for the size check, $2 \times 2 \times 4$ (64 sites, 2 756 394 ice states, ground sector 249 180, by Lanczos after reproducing every 48-site benchmark). Momentum assignments are certified by exact translation projectors.

Conclusion.—Treating the compactness of the emergent gauge field exactly changes what quantum spin ice is predicted to show. The free photon survives at long wavelength, where the field can be tipped gently. At the zone boundary, where it cannot, the response is dominated by a roton at half the free-photon energy, carrying half of the neutron weight and forming the lowest excitation of the model. The mechanism is Feynman’s, adapted to a saturated field: a wave that cannot be made by increasing E must be made by rearranging it, and the price is the ring-exchange coherence that rearrangement destroys. Alongside the roton the photon line splits and a multiparticle continuum appears, and an exact parity rule routes half the spectrum to non-magnetic probes. Seen from outside magnetism, these are exact strong-coupling spectra of a compact $U(1)$ gauge theory of the kind quantum simulators are being built to approximate [36, 37], obtained here from a gauge field that a magnet supplies for free.

Exact diagonalizations used the winding-free campaign infrastructure of Ref. [27].

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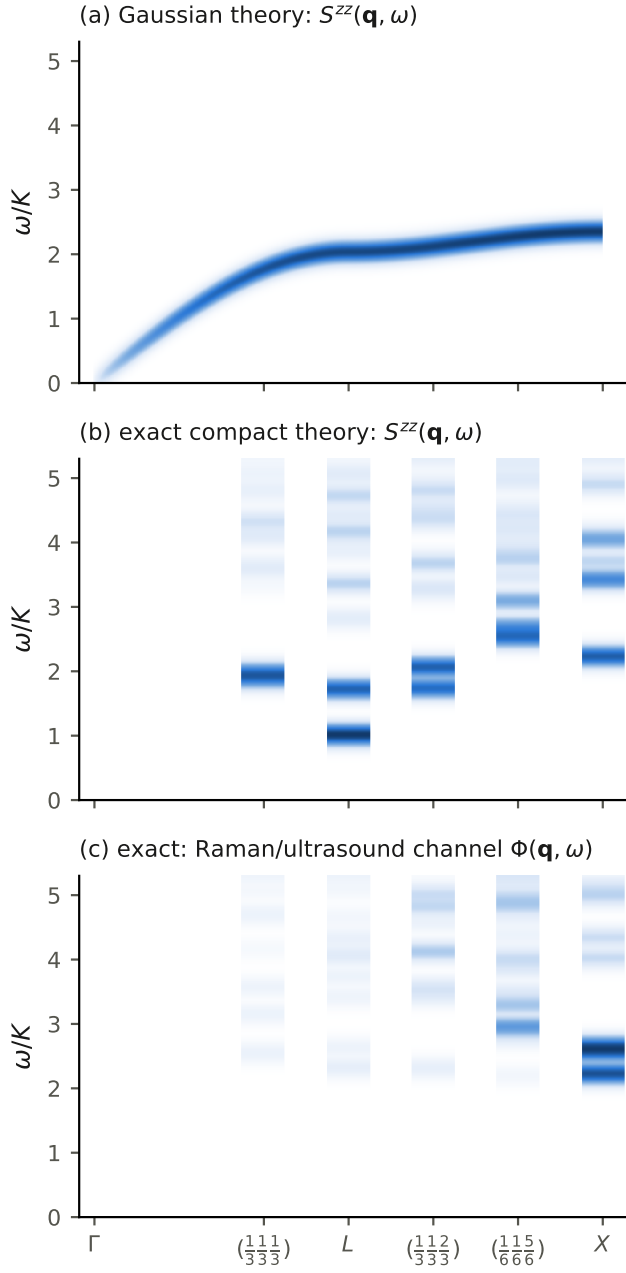


FIG. 3. Simulated measurements along Γ - L - X (48-site cluster, broadening $\eta = 0.08K$, each panel normalized to its own maximum). (a) The free-photon prediction for neutrons: one sharp line with all of the weight. (b) The exact compact theory: both polarizations, the roton dominating at L a factor of two below the free-photon line, and multiparticle weight above. (c) The spin-flip-even channel seen by Raman scattering and ultrasound, which by the selection rule shares no state with (a) or (b).